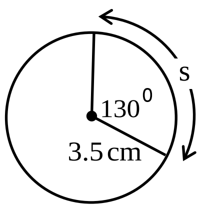
## Question 1 T1

Use the information in the diagram to determine the value of the arc length "s".




### Solution

$$130^0 \times \frac{\pi}{180^0} = \frac{13\pi}{18}$$

1 mark for conversion

$$s = \theta r$$

$$s = \frac{13\pi}{18}(3.5)$$

1 mark for substitution

**2 marks**

$$s = 7.941\ 248$$

$$s = 7.941 \text{ cm}$$

## Question 2 T5

Solve the following equation over the interval $[0,\ 2\pi)$.

$$\tan^2\theta + 2.8\tan\theta + 1.96 = 0$$

Use the quadratic formula $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ for $ax^2 + bx + c = 0.$

### Solution

$$\tan\theta = \frac{-2.8 \pm \sqrt{(2.8)^2 - 4(1)(1.96)}}{2(1)}$$ ½ mark for substitution

$$\tan\theta = \frac{-2.8 \pm 0}{2}$$

$$\tan\theta = -1.4$$ ½ mark for solving for $\tan\theta$

$$\theta_r = \tan^{-1}(1.4)$$
$$= 0.950\ 546$$
$$\theta = 2.191 \qquad \theta = 5.333$$ 1 mark (½ mark for each value of $\theta$)

**2 marks**

Determine how many monthly investments of $50 would have to be deposited into a savings account that pays 3% annual interest, compounded monthly, for the account's future value to be $50,000.

Use the formula:

$$\text{FV} = \frac{\text{R}\left[(1+i)^n - 1\right]}{i}$$

where FV = the future value

R = the investment amount

$$i = \frac{\text{the annual interest rate}}{\text{the number of compounding periods per year}}$$

$n$ = the number of investments

Express your answer as a whole number.

## Solution

$$50\ 000 = \frac{50\left[\left(1+\frac{0.03}{12}\right)^n - 1\right]}{\frac{0.03}{12}}$$

½ mark for substitution

$$50\ 000 = \frac{50\left[(1+0.0025)^n - 1\right]}{0.0025}$$

$$50\ 000 = 20\ 000\left(1.0025^n - 1\right)$$

$$2.5 = 1.0025^n - 1$$

$$3.5 = 1.0025^n$$

½ mark for simplification

$$\log 3.5 = \log 1.0025^n$$

½ mark for applying logarithms

$$\log 3.5 = n \log 1.0025$$

1 mark for power rule

$$n = \frac{\log 3.5}{\log 1.0025}$$

$$n = 501.73$$

½ mark for solving for $n$

∴ 502 monthly investments are needed.

**3 marks**

There are 5 men and 4 women to be seated in a row.

How many arrangements are possible if two men must sit at the beginning of the row and two men must sit at the end of the row?

### Solution

$$\underset{m}{\underline{5}} \cdot \underset{m}{\underline{4}} \cdot \underline{\quad 5! \quad} \cdot \underset{m}{\underline{3}} \cdot \underset{m}{\underline{2}} = 14\ 400$$

**or**

$$\underset{m}{\underline{5}} \cdot \underset{m}{\underline{4}} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1} \cdot \underset{m}{\underline{3}} \cdot \underset{m}{\underline{2}} = 14\ 400$$

1 mark for correctly restricting men at beginning and end of the row

1 mark for 5!

**2 marks**

a) In the binomial expansion of $\left(\frac{3}{x^2} - 4x^5\right)^8$, determine the 3rd term.

b) In the binomial expansion of $\left(\frac{3}{x^2} - 4x^5\right)^n$, the 6th term contains $x^{25}$.

Solve for $n$.

## Solution

a)

$$t_3 = {}_8C_2\left(\frac{3}{x^2}\right)^{8-2}\left(-4x^5\right)^2$$

2 marks (1 mark for ${}_8C_2$, ½ mark for each consistent factor)

$$t_3 = 28\left(\frac{3^6}{x^{12}}\right)\left(\frac{16x^{10}}{1}\right)$$

$$= \frac{28}{1} \times \frac{729}{x^{12}} \times \frac{16x^{10}}{1}$$

1 mark for simplification (½ mark for coefficient, ½ mark for exponents)

$$= \frac{326\ 592}{x^2}$$

**3 marks**

b)

$$x^{25} = \left(x^{-2}\right)^{n-5}\left(x^5\right)^5$$

1 mark for substitution

$$x^{25} = x^{-2n+10+25}$$

$$25 = -2n + 35$$

1 mark for equating exponents

$$-10 = -2n$$

$$5 = n$$

**2 marks**

## Question 6 R1

Given the following two functions, $f(x) = \sqrt{x-1}$ and $g(x) = x^2 + 1$, evaluate $g(f(3))$.

### Solution

#### Method 1

$$\begin{aligned} f(3) &= \sqrt{3-1} \\ &= \sqrt{2} \end{aligned}$$

½ mark for $f(3)$

$$\begin{aligned} g\left(\sqrt{2}\right) &= \left(\sqrt{2}\right)^2 + 1 \\ &= 2+1 \\ &= 3 \end{aligned}$$

½ mark for consistent value of $g(f(3))$

**1 mark**

#### Method 2

$$g(f(x)) = \left(\sqrt{x-1}\right)^2 + 1$$

½ mark for $g(f(x))$

$$\begin{aligned} &= x-1+1 \\ &= x \end{aligned}$$

$$g(f(3)) = 3$$

½ mark for evaluating $g(f(3))$

**1 mark**

## Question 7 T2

If $\theta$ terminates in quadrant II and $\csc\theta = \frac{3}{2}$, determine the exact value of $\tan\theta$.

### Solution

$$\csc\theta = \frac{r}{y}$$

$$x^2 + y^2 = r^2$$

$$x^2 + 2^2 = 3^2$$ ½ mark for identifying $y = 2,\ r = 3$

$$x^2 = 5$$

$$x = \pm\sqrt{5}$$ ½ mark for solving for $x$

$$\tan\theta = -\frac{2}{\sqrt{5}}$$ 1 mark for $\tan\theta$ (½ mark for quadrant, ½ mark for value)

**2 marks**

Note(s):

§ accept any of the following values for *x*: $x = \pm\sqrt{5}$, $x = \sqrt{5}$, or $x = -\sqrt{5}$

## Question 8 R11

a) Determine the remainder when $x^4 - 3x^2 + 1$ is divided by $x + 2$.

b) Is $x + 2$ a factor of $x^4 - 3x^2 + 1$? Explain your reasoning.

### Solution

a) $(-2)^4 - 3(-2)^2 + 1$

$16 - 12 + 1$

$5$

1 mark for remainder theorem

**or**

$$\begin{array}{r|rrrrr} -2 & 1 & 0 & -3 & 0 & 1 \\ & & -2 & 4 & -2 & 4 \\ \hline & 1 & -2 & 1 & -2 & 5 \end{array}$$

**or**

1 mark for synthetic division

**1 mark**

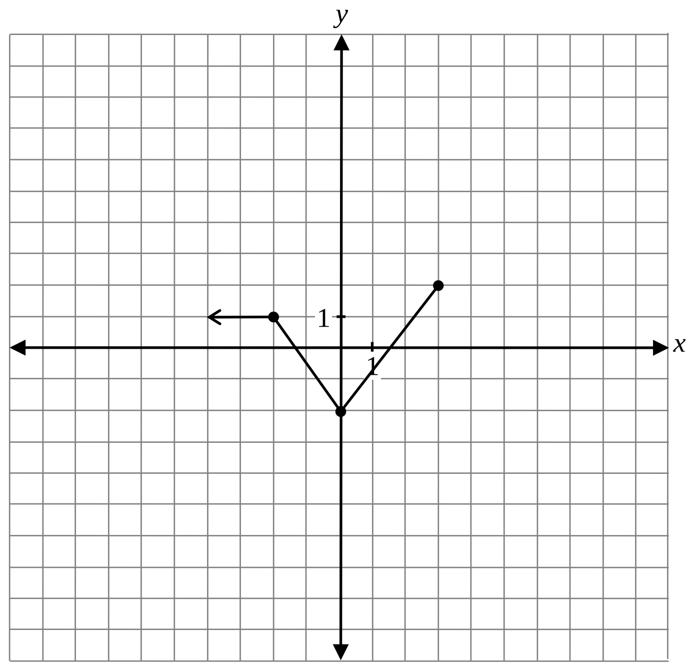
The remainder is 5.

b) For $x + 2$ to be a factor of $x^4 - 3x^2 + 1$, the remainder must be 0. Since the remainder is 5, $x + 2$ is not a factor.

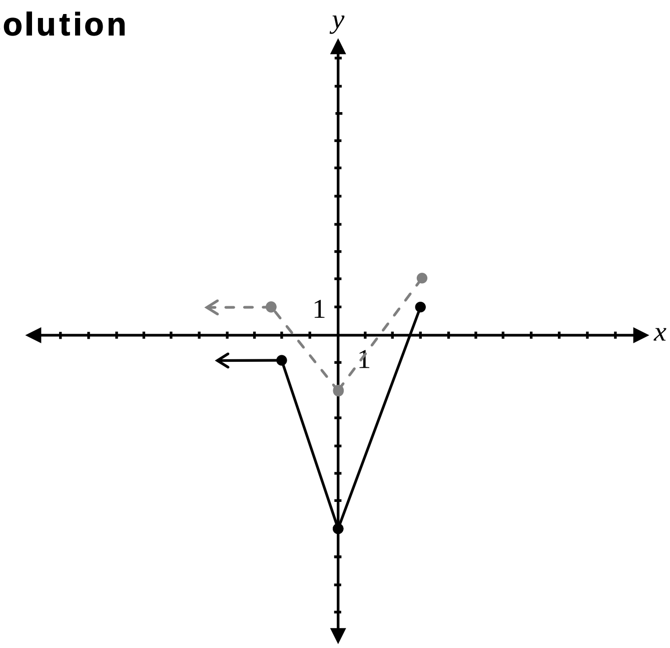
1 mark for explanation

**1 mark**

Given the graph of $y = f(x)$ below, sketch the graph of $y = 2f(x) - 3$.




**Solution**




1 mark for vertical stretch
1 mark for vertical shift

**2 marks**

## Question 10 R6

Determine one possible restriction for the domain of $f(x)=(x-1)^2$ so that the inverse of $f(x)$ is a function.

### Solution

$x \geq 1$

**or**

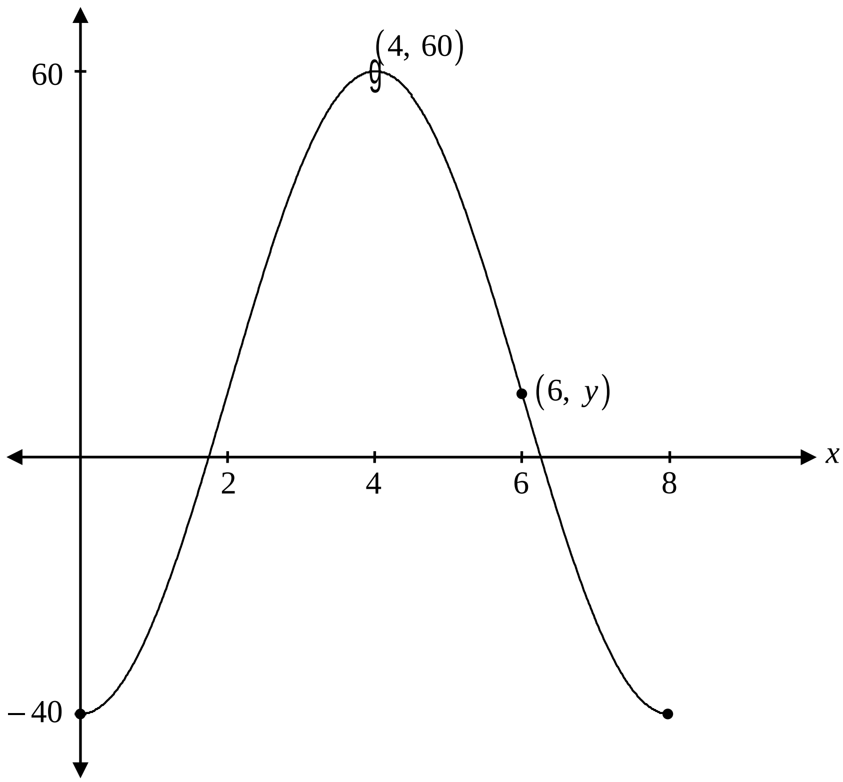
$x < 1$

**1 mark**

Note(s):

§ Many solutions are possible. Any solutions which restrict $f(x)$ to a one-to-one function are correct.

Using the graph of the sinusoidal function below, find the value of $y$ in the point $(6,\ y)$.




**Solution**

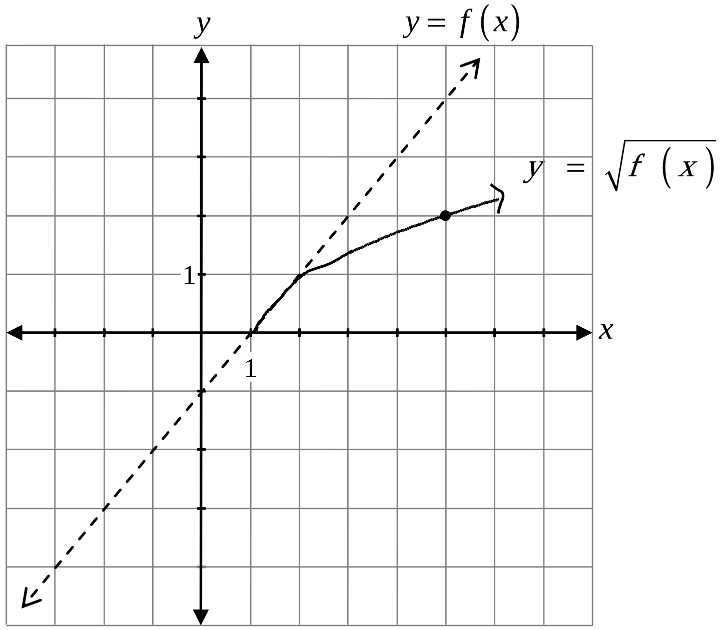
$y = 10$

**1 mark**

Billy was given the graph of $y = f(x)$.

He was asked to sketch the graph of $y = \sqrt{f(x)}$.

His answer is given on the graph below.




Explain the error Billy made when sketching the graph of $y = \sqrt{f(x)}$.

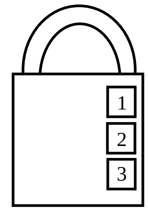
## Solution

Billy's graph should be above the line $y = f(x)$ over the interval where $x$ is between 1 and 2.

**1 mark**

## Question 13 P2, P3

Explain why a locker combination should really be called a locker permutation.




### Solution

The order of the locker combination matters; therefore, it is a permutation.

**1 mark**

## Question 14 R5

The graph of $f(x) = x^2 + 4$ is reflected over the $x$-axis.
Write the equation of the new function.

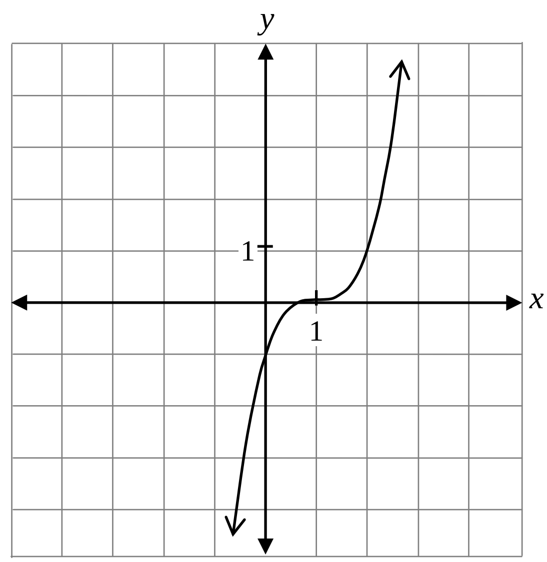
$y =$ ______________________

### Solution

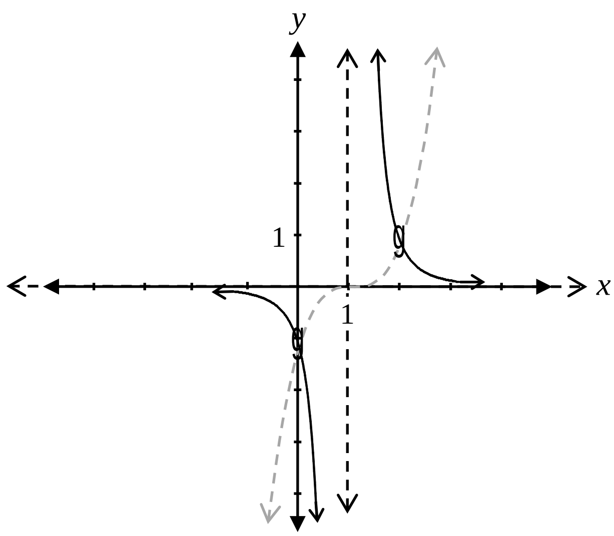
$y = -x^2 - 4$

**1 mark**

Given the graph of $y = f(x)$ below, sketch the graph of $y = \dfrac{1}{f(x)}$.




**Solution**




1 mark for vertical asymptote at $x = 1$
½ mark for graph left of vertical asymptote
½ mark for graph right of vertical asymptote

**2 marks**

Divide $\left(x^3 - 5x - 4\right)$ by $(x + 1)$.

## Solution

### Method 1

$$\begin{array}{r|rrrr} -1 & 1 & 0 & -5 & -4 \\ & & -1 & 1 & 4 \\ \hline & 1 & -1 & -4 & 0 \end{array}$$

1 mark for set-up of synthetic division using addition

$x^2 - x - 4$

1 mark for quotient

**2 marks**

### Method 2

$$\begin{array}{r}
x^2 - x - 4 \\
x+1\overline{\big)\,x^3 + 0x^2 - 5x - 4} \\
\underline{-\left(x^3 + x^2\right)} \qquad\qquad \\
-x^2 - 5x \qquad \\
\underline{-\left(-x^2 - x\right)} \qquad \\
-4x - 4 \\
\underline{-\left(-4x - 4\right)} \\
0
\end{array}$$

1 mark for set-up of long division
1 mark for quotient

**2 marks**

### Method 3

$$\begin{array}{r|rrrr} 1 & 1 & 0 & -5 & -4 \\ & & 1 & -1 & -4 \\ \hline & 1 & -1 & -4 & 0 \end{array}$$

1 mark for set-up of synthetic division using subtraction

$x^2 - x - 4$

1 mark for quotient

**2 marks**

## Question 17 P4

You are given the following row of Pascal's Triangle.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 7 | 21 | 35 | 35 | 21 | 7 | 1 |

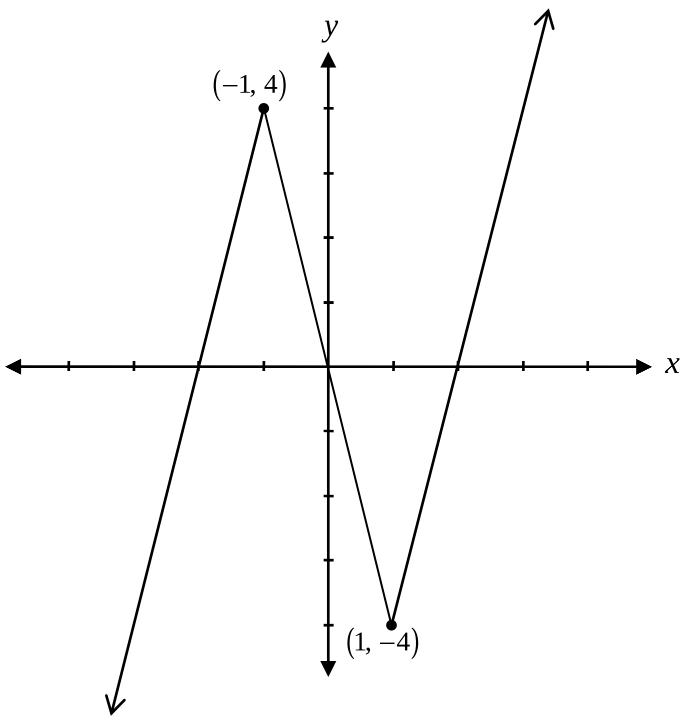
Determine the values of the next row.

### Solution

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | 8 | 28 | 56 | 70 | 56 | 28 | 8 | 1 |

**1 mark**

Given the graph of $y = f(x)$ below, state the domain and range of $y = \sqrt{f(x)}$.




## Solution

Domain: $\{x \in \mathbb{R} \mid -2 \le x \le 0 \text{ or } x \ge 2\}$

**or**

$[-2, 0] \cup [2, \infty)$

1 mark for domain (½ mark for $-2 \le x \le 0$, ½ mark for $x \ge 2$)

Range: $\{y \in \mathbb{R} \mid 0 \le y\}$

**or**

$[0, \infty)$

1 mark for range

**2 marks**

## Question 19 T6

Prove the identity below for all permissible values of $\theta$:

$$\frac{1-\tan^2\theta}{1+\tan^2\theta}=\cos 2\theta$$

## Solution

### Method 1

| Left-Hand Side | Right-Hand Side |
|---|---|
| $\dfrac{1-\tan^2\theta}{1+\tan^2\theta}$ | $\cos 2\theta$ |
| $\dfrac{1-\tan^2\theta}{\sec^2\theta}$ | |
| $\dfrac{1}{\sec^2\theta}-\dfrac{\tan^2\theta}{\sec^2\theta}$ | |
| $\cos^2\theta-\dfrac{\frac{\sin^2\theta}{\cos^2\theta}}{\frac{1}{\cos^2\theta}}$ | |
| $\cos^2\theta-\sin^2\theta$ | |
| $\cos 2\theta$ | |

1 mark for correct substitution of appropriate identities
1 mark for algebraic strategies
1 mark for logical process to prove the identity

**3 marks**

**Method 2**

| Left-Hand Side | Right-Hand Side |
|---|---|
| $\dfrac{1-\tan^2\theta}{1+\tan^2\theta}$ | $\cos 2\theta$ |
| $\dfrac{1-\dfrac{\sin^2\theta}{\cos^2\theta}}{1+\dfrac{\sin^2\theta}{\cos^2\theta}}$ | |
| $\dfrac{\dfrac{\cos^2\theta-\sin^2\theta}{\cos^2\theta}}{\dfrac{\cos^2\theta+\sin^2\theta}{\cos^2\theta}}$ | |
| $\dfrac{\cos^2\theta-\sin^2\theta}{\cos^2\theta+\sin^2\theta}$ | |
| $\cos^2\theta-\sin^2\theta$ | |
| $\cos 2\theta$ | |

1 mark for correct substitution of appropriate identities
1 mark for algebraic strategies
1 mark for logical process to prove the identity

**3 marks**

**Method 3**

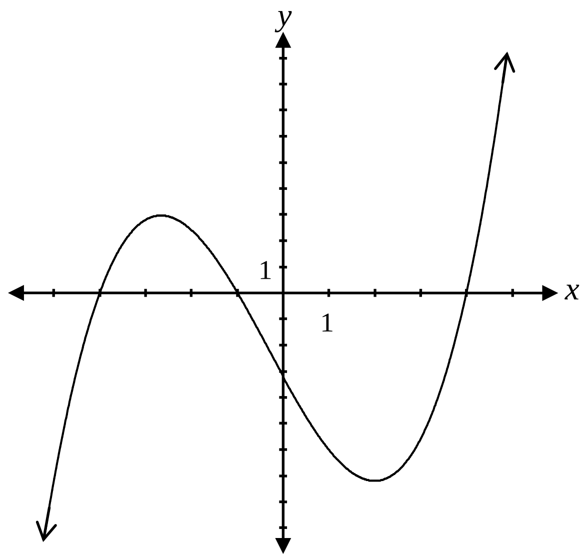
| Left-Hand Side | Right-Hand Side |
|---|---|
| $\dfrac{1-\tan^2\theta}{1+\tan^2\theta}$ | $\cos 2\theta$ |
| $\dfrac{1-\left(\sec^2\theta-1\right)}{\sec^2\theta}$ | |
| $\dfrac{2-\sec^2\theta}{\sec^2\theta}$ | |
| $\dfrac{2-\dfrac{1}{\cos^2\theta}}{\dfrac{1}{\cos^2\theta}}$ | |
| $\dfrac{\dfrac{2\cos^2\theta-1}{\cos^2\theta}}{\dfrac{1}{\cos^2\theta}}$ | |
| $2\cos^2\theta-1$ | |
| $\cos 2\theta$ | |

1 mark for correct substitution of appropriate identities
1 mark for algebraic strategies
1 mark for logical process to prove the identity

**3 marks**

## Question 20 R1

Given the graph of the function of $f(x)$ below, what is the range of $y = |f(x)|$?




a) $y \in$ ¡

b) $y \geq -7$

c) $y \geq 0$

d) $-4 \leq y \leq -1$ or $y \geq 4$

## Question 21 R8

Simplify the following expression:

$$\frac{1}{2}\log_a 36 - \log_a 2$$

a) $\log_a 3$

b) $\log_a 4$

c) $\log_a 9$

d) $\log_a 12$

## Question 22 R2

Given $f(x) = x^2 - x + 2$, an equation that represents the graph of $f(x)$ shifted 3 units to the right is:

a) $y = (x+3)^2 - (x+3) - 3$

b) $y = (x-3)^2 - (x-3) + 2$

c) $y = (x-3)^2 - x - 2$

d) $y = x^2 - x + 2 - 3$

## Question 23 R13

What is the domain of the function $y = \sqrt{-4x}$?

a) $\{x \in \mathbb{R} \mid x \geq 2\}$

b) $\{x \in \mathbb{R} \mid x \leq 2\}$

c) $\{x \in \mathbb{R} \mid x \geq 0\}$

d) $\{x \in \mathbb{R} \mid x \leq 0\}$

## Question 24 R14

Which of the following is true about the two functions below?

$$f(x) = \frac{(x+2)(x-2)}{x-2} \qquad g(x) = \frac{(x-2)(x+1)}{(x+2)(x-2)}$$

a) Both have a point of discontinuity (hole) when $x = 2$.

b) Both have the same vertical asymptote.

c) Both have the same horizontal asymptote.

d) Both have the same $y$-intercept.

## Question 25 T5

The general solution to the equation $\cos\theta = -\frac{1}{2}$ is:

a) $\left.\begin{aligned} \theta &= \frac{\pi}{3} + 2\pi k \\ \theta &= \frac{5\pi}{3} + 2\pi k \end{aligned}\right\}$ where $k \in \mathrm{I}$

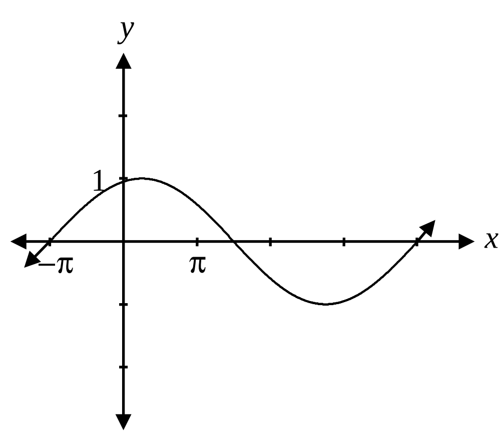
b) $\left.\begin{aligned} \theta &= \frac{\pi}{3} + \pi k \\ \theta &= \frac{5\pi}{3} + \pi k \end{aligned}\right\}$ where $k \in \mathrm{I}$

c) $\left.\begin{aligned} \theta &= \frac{2\pi}{3} + 2\pi k \\ \theta &= \frac{4\pi}{3} + 2\pi k \end{aligned}\right\}$ where $k \in \mathrm{I}$

d) $\left.\begin{aligned} \theta &= \frac{2\pi}{3} + \pi k \\ \theta &= \frac{4\pi}{3} + \pi k \end{aligned}\right\}$ where $k \in \mathrm{I}$

## Question 26 T4

If the equation $y = \sin\left(b\left(x + \pi\right)\right)$ is represented by the following graph, what is the value of $b$?




a) $\frac{2}{5}$

b) $\frac{5}{2}$

c) $\frac{2\pi}{5}$

d) $5\pi$

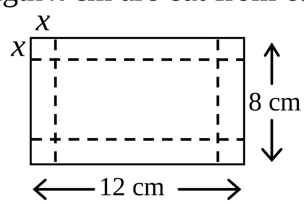
## Question 27 R7

Which of the following is closest to the value of $\log_2 40 + \log_5 125$?

a) 3 b) 8 c) 10 d) 45

## Question 28 R12

A sheet of paper 12 cm long and 8 cm wide is used to make a box with no lid. Equal squares of side length *x* cm are cut from each of the corners and the sides are folded up to make the box.




Which of the following expresses the volume of the box?

a) $V(x) = x(12 + x)(8 + x)$

b) $V(x) = x(12 - x)(8 - x)$

c) $V(x) = x(12 + 2x)(8 + 2x)$

d) $V(x) = x(12 - 2x)(8 - 2x)$

## Question 29 R5

Given that the graph of $f(x)$ contains the point $(-3, 5)$, what point must be on the graph of $f(-x)$?

a) $(-3, -5)$

b) $(3, 5)$

c) $(3, -5)$

d) $(5, -3)$

## Question 30 T1

Determine one positive and one negative coterminal angle with the angle $\frac{5\pi}{6}$.

### Solution

$\frac{17\pi}{6}$ and $-\frac{7\pi}{6}$

½ mark for one positive coterminal angle
½ mark for one negative coterminal angle

**1 mark**

Note(s):

§ Other answers are possible.
§ Answers in degrees are acceptable.

## Question 31 T3

Evaluate:

$$\left(\sin\frac{11\pi}{3}\right)\left(\sec\frac{11\pi}{6}\right)$$

### Solution

$$\left(-\frac{\sqrt{3}}{2}\right)\left(\frac{2}{\sqrt{3}}\right)=-1$$

1 mark for $\sin\frac{11\pi}{3}$ (½ mark for quadrant, ½ mark for value)

1 mark for $\sec\frac{11\pi}{6}$ (½ mark for quadrant, ½ mark for value)

**2 marks**

## Question 32 T5

Given the equation $2\sin^2\theta - 3\sin\theta + 1 = 0$, verify that $\theta = \frac{\pi}{2}$ is a solution.

### Solution

$$\begin{aligned} \text{Left-hand side } &= 2\left(\sin\frac{\pi}{2}\right)^2 - 3\left(\sin\frac{\pi}{2}\right) + 1 \\ &= 2(1)^2 - 3(1) + 1 \\ &= 0 \\ &= \text{ Right-hand side} \end{aligned}$$

1 mark for verification

**1 mark**

Using the laws of logarithms, expand:

$$\log_a \left( \frac{x \cdot y}{z} \right)$$

## Solution

$$\log_a x + \log_a y - \log_a z$$

1 mark for product rule
1 mark for quotient rule

**2 marks**

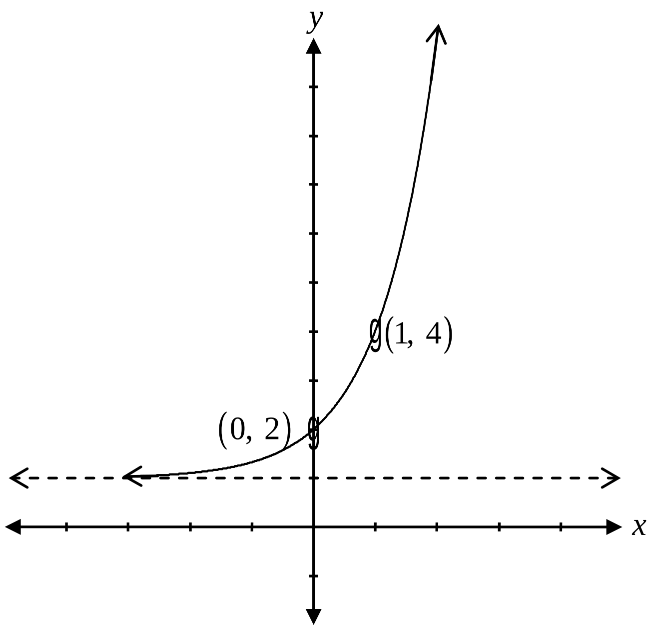
## Question 34 a) R9 b) R6

a) Sketch the graph of $f(x) = 3^x + 1$.

b) Sketch the graph of $f^{-1}(x)$.

### Solution

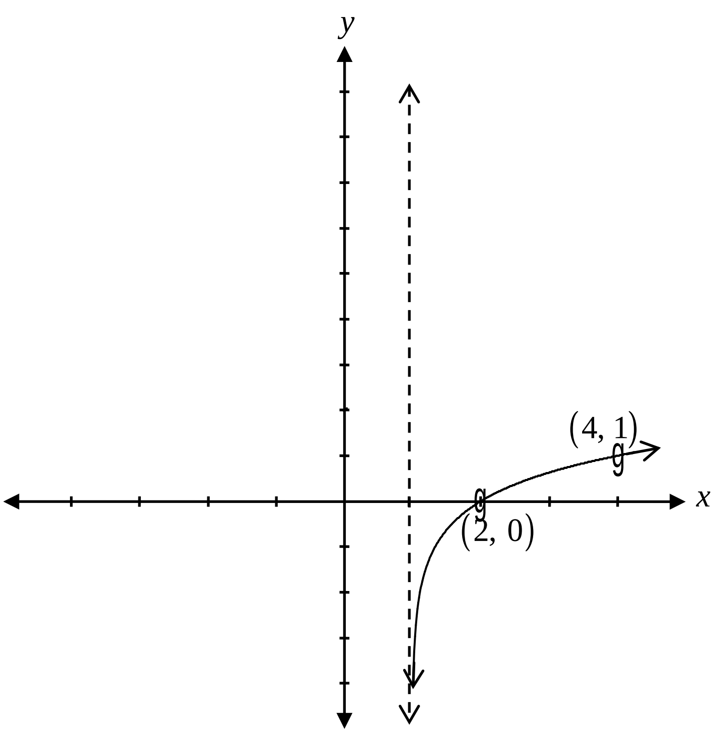
a)




½ mark for increasing exponential function
½ mark for $y$-intercept at $(0, 2)$
½ mark for asymptote at $y = 1$
½ mark for consistent point on exponential function

**2 marks**

b)




1 mark for consistent graph of the inverse

**1 mark**

## Question 35 R9

Determine the $x$-intercept and $y$-intercept of $y = \log_2(x+4) - 1$.

### Solution

Substitute $x$ with 0.

$y = \log_2 4 - 1$

$y = 2 - 1$ ½ mark for evaluating logarithm

$y = 1$ ½ mark for consistent value of $y$

$\therefore$ $y$-intercept is 1

Substitute $y$ with 0.

$0 = \log_2(x + 4) - 1$

$1 = \log_2(x + 4)$

$2 = x + 4$ ½ mark for exponential form

$-2 = x$ ½ mark for consistent value of $x$

$\therefore$ $x$-intercept is $-2$

**2 marks**

Note(s):

§ award ½ mark if student substitutes $x$ with 0 to find the $y$-intercept and $y$ with 0 to find the $x$-intercept

Explain the error that was made when solving the following equation:

$$\sin 2\theta = \cos\theta, \text{ where } \theta \in \mathbb{R}$$

$$\sin 2\theta = \cos\theta$$
$$2\sin\theta\cos\theta = \cos\theta$$
$$\frac{2\sin\theta\cos\theta}{\cos\theta} = \frac{\cos\theta}{\cos\theta}$$
$$2\sin\theta = 1$$
$$\sin\theta = \frac{1}{2}$$
$$\theta = \frac{\pi}{6} + 2k\pi,\ \frac{5\pi}{6} + 2k\pi,\ k \in I$$

### Solution

The student divided by $\cos\theta$ instead of factoring out $\cos\theta$.

**1 mark**

**or**

There are two more solutions that come from the equation $\cos\theta = 0$.

**or**

The student cannot divide both sides by $\cos\theta$ since $\cos\theta$ could equal 0.

Given $f(x) = x^2 - 2x - 3$ and $g(x) = x + 1$:

a) Write the equation of $y = f(g(x))$.

b) Sketch the graph of $y = f(g(x))$.

**Solution**

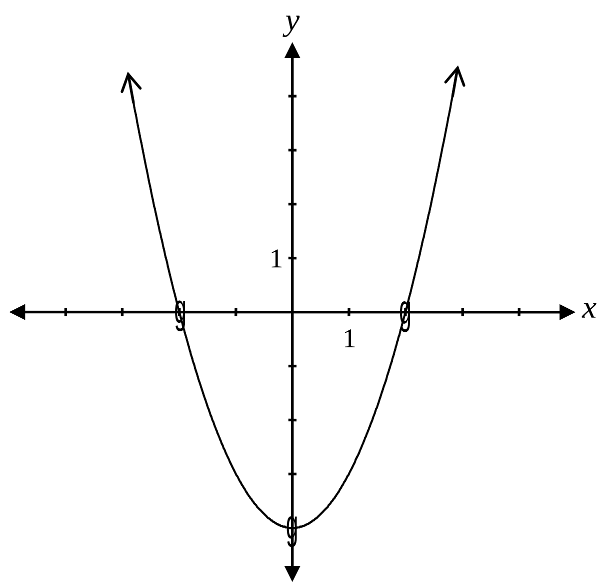
a)

$$f(g(x)) = (x+1)^2 - 2(x+1) - 3$$
$$= x^2 + 2x + 1 - 2x - 2 - 3$$
$$= x^2 - 4$$

1 mark for composition

**1 mark**

b)




1 mark for graph (½ mark for *x*-intercepts, ½ mark for *y*-intercept)

**1 mark**

## Question 38 T2

Is the point $\left(\frac{3}{4}, -\frac{\sqrt{3}}{4}\right)$ on the unit circle?

Justify your answer.

### Solution

$$x^2 + y^2 = 1$$

$$\text{Left-hand side } = \left(\frac{3}{4}\right)^2 + \left(-\frac{\sqrt{3}}{4}\right)^2$$

½ mark for substitution

$$= \frac{9}{16} + \frac{3}{16}$$

$$= \frac{12}{16}$$

$$\frac{12}{16} \neq 1 \;\therefore\; \text{not on unit circle}$$

½ mark for justification

**1 mark**

## Question 39 T3

Explain why the equation $\sec\theta = \frac{1}{4}$ has no solution.

### Solution

The value of $\sec\theta$ cannot be between $-1$ and 1.

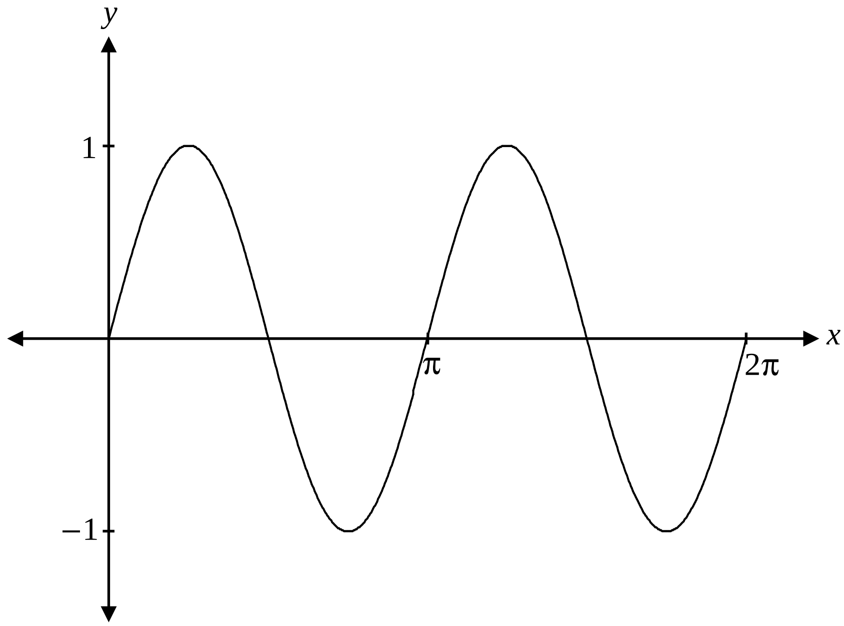
**1 mark**

**or**

$\cos\theta$ cannot be greater than 1.

The graph of $y = \sin 2x$ is sketched below.

Explain how to use this graph to solve the equation $\sin 2x = \frac{1}{2}$ over the interval $[0, 2\pi]$.



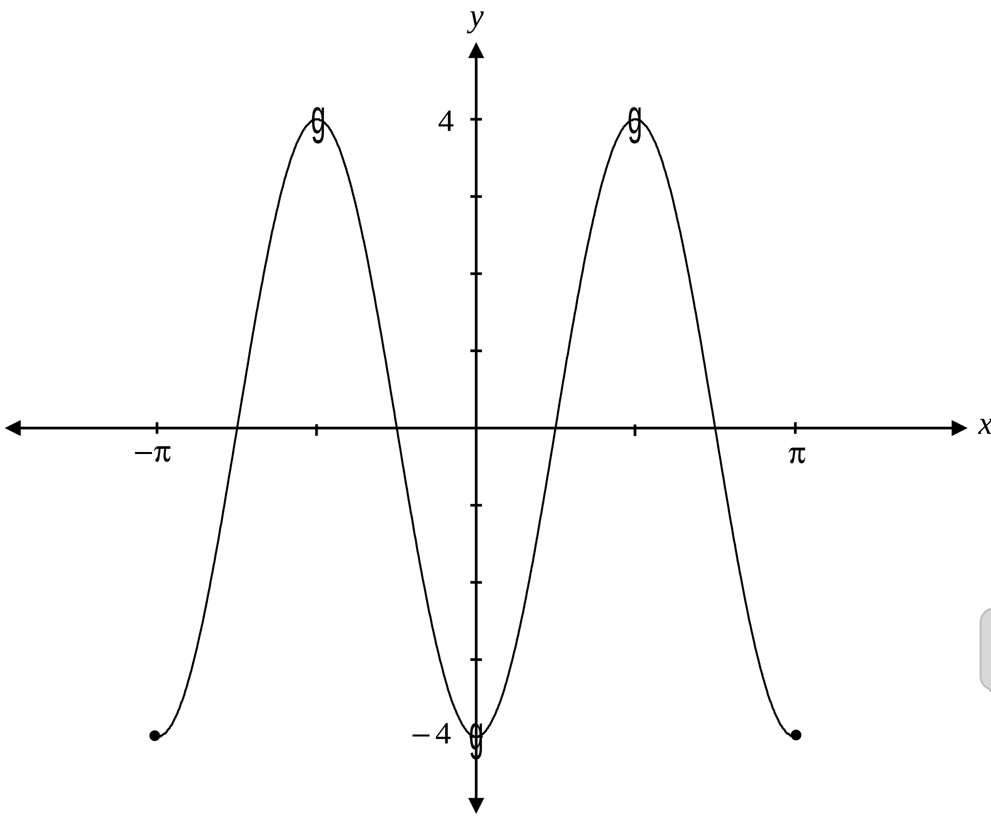

**Solution**

Draw the line $y = \frac{1}{2}$. The solution will be the $x$-values where the two graphs intersect.

**1 mark**

Sketch the graph of $y = -4\cos(2x)$ over the interval $[-\pi, \pi]$.

**Solution**




1 mark for vertical reflection
1 mark for range
1 mark for period

**3 marks**

Write the equation for $f(x)$ that satisfies all of the following conditions:

- $f(x)$ is a polynomial function of degree 4
- $f(x)$ has a zero at 2 with a multiplicity of 3
- $f(x)$ has a zero at $-5$
- $f(x)$ has a $y$-intercept of 80

**Solution**

$$f(x) = a(x-2)^3(x+5)$$
$$80 = a(0-2)^3(0+5)$$
$$80 = a(-8)(5)$$
$$80 = -40a$$
$$a = -2$$

$$\therefore\ f(x) = -2(x-2)^3(x+5)$$

½ mark for the factors $(x-2)$ and $(x+5)$
½ mark for multiplicity of 3
½ mark for substitution/negative value of *a*

½ mark for value of 2 for *a*

**2 marks**

## Question 43 T6

Find the exact value of $\sin\left(\frac{19\pi}{12}\right)$.

### Solution

$$\sin\left(\frac{10\pi}{12}+\frac{9\pi}{12}\right)=\sin\left(\frac{5\pi}{6}+\frac{3\pi}{4}\right)$$

1 mark for combination

$$\sin\frac{19\pi}{12}=\sin\frac{5\pi}{6}\cos\frac{3\pi}{4}+\cos\frac{5\pi}{6}\sin\frac{3\pi}{4}$$

$$=\left(\frac{1}{2}\right)\left(-\frac{\sqrt{2}}{2}\right)+\left(-\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right)$$

2 marks (½ mark for each exact value)

**3 marks**

$$=-\frac{\sqrt{2}}{4}-\frac{\sqrt{6}}{4}$$

$$=\frac{-\sqrt{2}-\sqrt{6}}{4}$$

Note(s):

§ Other combinations are possible.

## Question 44 R10

Solve the following equation:

$$2\log_2(x-1)-\log_2(x-5)=\log_2(x+1)$$

## Solution

### Method 1

$\log_2 \dfrac{(x-1)^2}{x-5}=\log_2(x+1)$ — 2 marks for logarithmic rules (1 mark for power rule, 1 mark for quotient rule)

$\dfrac{(x-1)^2}{x-5}=x+1$ — 1 mark for equating arguments

$x^2-2x+1=x^2-4x-5$

$2x=-6$

$\cancel{x=-3}$ — ½ mark for solving for $x$

$\therefore$ no solution — ½ mark for no solution

**4 marks**

### Method 2

$2\log_2(x-1)=\log_2(x+1)+\log_2(x-5)$

$\log_2(x-1)^2=\log_2(x+1)(x-5)$ — 2 marks for logarithmic rules (1 mark for power rule, 1 mark for product rule)

$(x-1)^2=(x+1)(x-5)$ — 1 mark for equating arguments

$x^2-2x+1=x^2-4x-5$

$2x=-6$

$\cancel{x=-3}$ — ½ mark for solving for $x$

$\therefore$ no solution — ½ mark for no solution

**4 marks**

**Method 3**

$$\log_2 (x-1)^2 - \log_2 (x-5) - \log_2 (x+1) = 0$$

$$\log_2 \frac{(x-1)^2}{(x-5)(x+1)} = 0$$

2 marks for logarithmic rules (1 mark for power rule, 1 mark for quotient rule)

$$2^0 = \frac{(x-1)^2}{(x-5)(x+1)}$$

1 mark for exponential form

$$x^2 - 4x - 5 = x^2 - 2x + 1$$

$$-6 = 2x$$

$$\cancel{-3 = x}$$

½ mark for solving for $x$

$\therefore$ no solution

½ mark for no solution

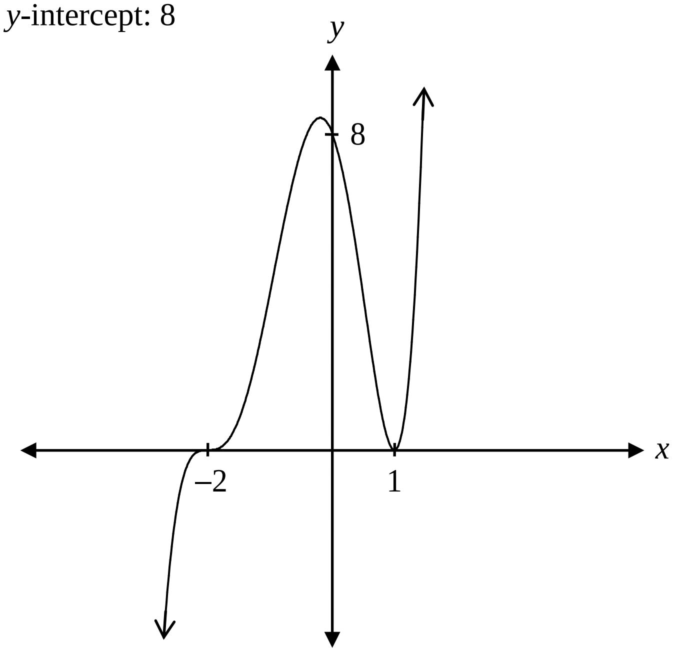
**4 marks**

Sketch the graph of $f(x) = (x-1)^2 (x+2)^3$.

Label the $x$-intercepts and the $y$-intercept.

## Solution

$x$-intercepts: $-2, 1$
$y$-intercept: 8




1 mark for $x$-intercepts

½ mark for $y$-intercept

1 mark for multiplicity (½ mark for degree of 2, ½ mark for degree of 3)

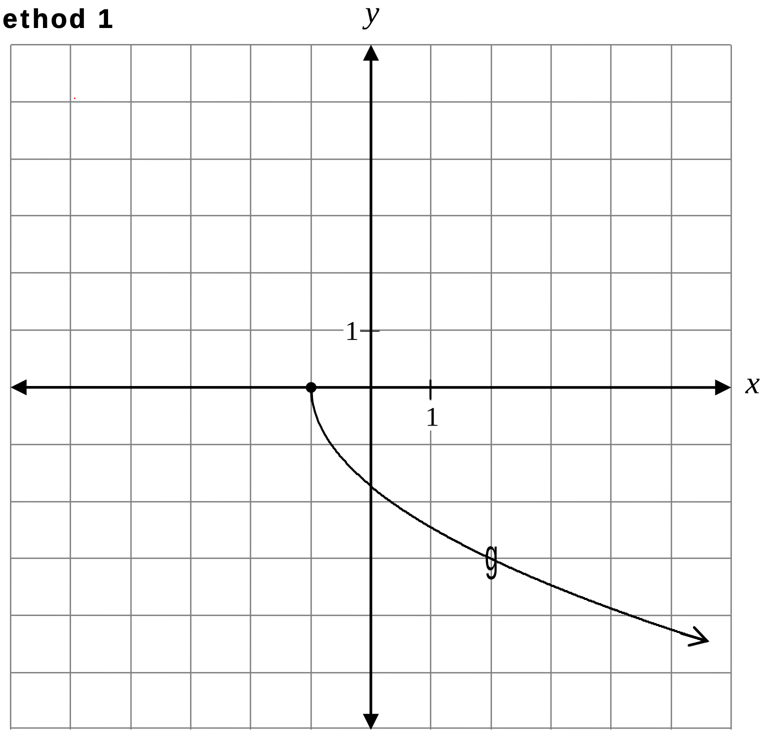
½ mark for end behaviour

**3 marks**

Sketch the graph of $y = -\sqrt{3(x+1)}$.

## Solution

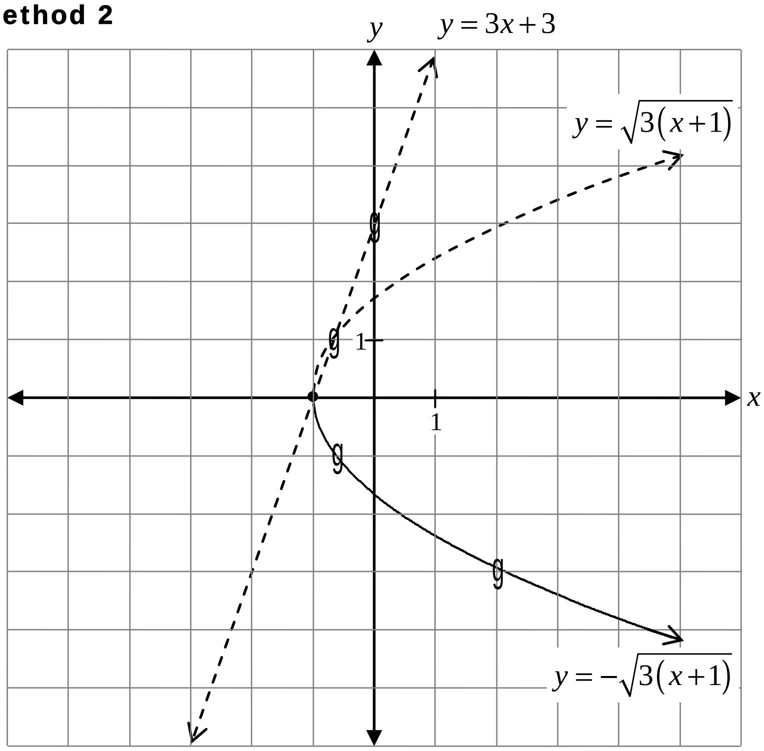
### Method 1




1 mark for horizontal shift
1 mark for vertical reflection
1 mark for shape (graph of a radical function)
1 mark for horizontal compression

**4 marks**

### Method 2




1 mark for invariant point where $y = 0$ and $y = 1$ (½ mark for each point)
1 mark for domain of $y = -\sqrt{3(x+1)}$: $[-1, \infty)$
1 mark for reflection in the $x$-axis
½ mark for shape between invariant points
½ mark for shape to the right of the invariant points

**4 marks**

Solve:

$${}_{n-1}P_2 = 42$$

## Solution

$$\frac{(n-1)!}{(n-1-2)!} = 42$$ ½ mark for substitution

$$\frac{(n-1)(n-2)(n-3)!}{(n-3)!} = 42$$ 1 mark for factorial expansion

$$(n-1)(n-2) = 42$$ ½ mark for simplification of factorials

$$n^2 - 3n + 2 = 42$$

$$n^2 - 3n - 40 = 0$$

$$(n-8)(n+5) = 0$$

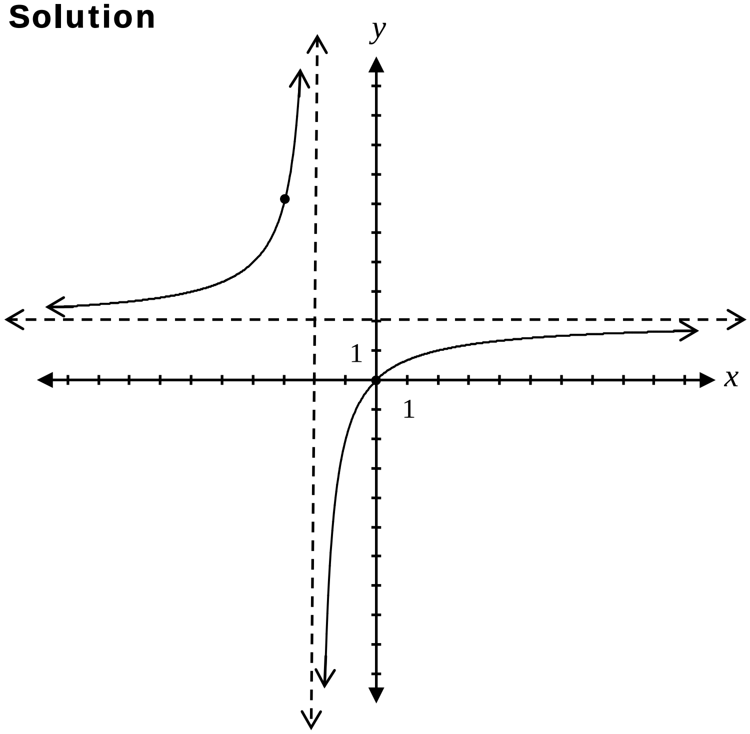
$$n = 8 \quad \cancel{n = -5}$$

½ mark for solving for $n$
½ mark for rejecting extraneous root

**3 marks**

Sketch the graph of $y = \dfrac{2x}{x+2}$.

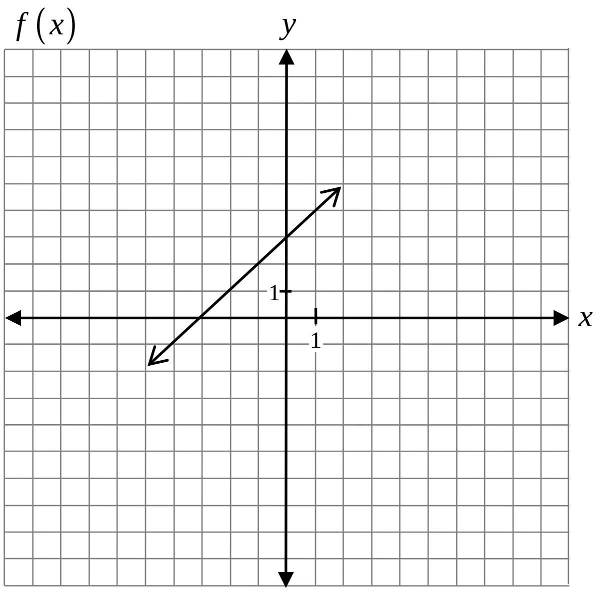
**Solution**

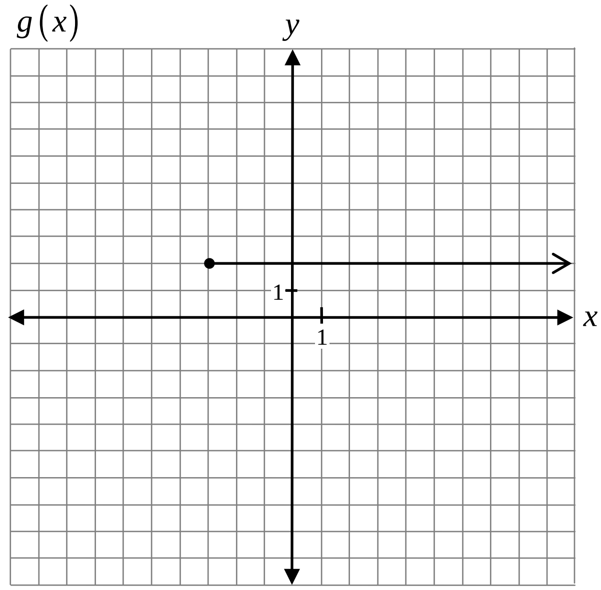



1 mark for horizontal asymptote at $y = 2$
1 mark for vertical asymptote at $x = -2$
½ mark for graph left of vertical asymptote
½ mark for graph right of vertical asymptote

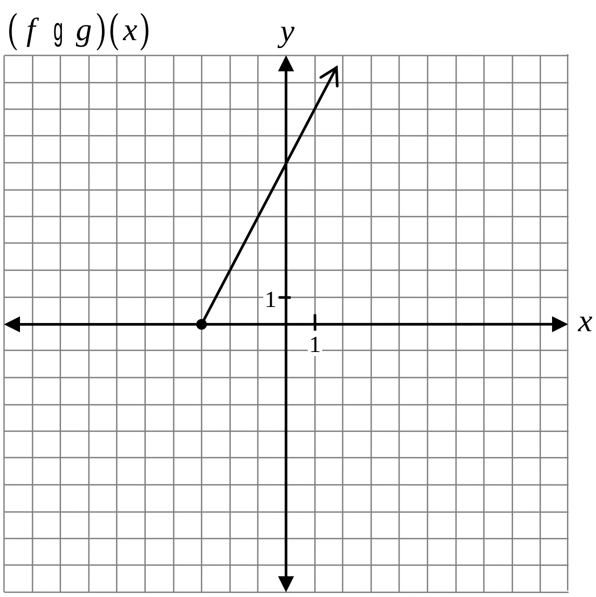
**3 marks**

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $(f \cdot g)(x)$.




## Solution




1 mark for operation of multiplication
1 mark for restricted domain

**2 marks**